

Le problème des missionnaires et des cannibales

EXTENDS *Naturals, FiniteSets*

CONSTANTS *numMiss, numCan*

ASSUME $numMiss \in Nat \wedge numCan \in Nat$

VARIABLES

missG,
missD,
canG,
canD,
barque

$TypeOK \triangleq \square(barque \in \{ "G", "D" \})$

$NombreOK \triangleq \square($
 $\quad \wedge missG + missD = numMiss$
 $\quad \wedge canG + canD = numCan$
 $)$

$pasMiam \triangleq$
 $\quad \wedge missG > 0 \Rightarrow missG \geq canG$
 $\quad \wedge missD > 0 \Rightarrow missD \geq canD$

$ToujoursOK \triangleq \square pasMiam$

$Solution \triangleq$
 $\quad \square \neg (missD = numMiss \wedge canD = numCan)$

prédicats sur la *barque*

$barqueGD \triangleq$
 $\quad \wedge barque = "G"$
 $\quad \wedge barque' = "D"$

$barqueDG \triangleq$
 $\quad \wedge barque = "D"$
 $\quad \wedge barque' = "G"$

prédicats sur les missionnaires

$missGD1 \triangleq$
 $\quad \wedge missG' = missG - 1$
 $\quad \wedge missD' = missD + 1$

$missGD2 \triangleq$

$$\begin{aligned} &\wedge \text{miss}G' = \text{miss}G - 2 \\ &\wedge \text{miss}D' = \text{miss}D + 2 \end{aligned}$$

$$\begin{aligned} \text{miss}DG1 &\triangleq \\ &\wedge \text{miss}D' = \text{miss}D - 1 \\ &\wedge \text{miss}G' = \text{miss}G + 1 \end{aligned}$$

$$\begin{aligned} \text{miss}DG2 &\triangleq \\ &\wedge \text{miss}D' = \text{miss}D - 2 \\ &\wedge \text{miss}G' = \text{miss}G + 2 \end{aligned}$$

prédicats sur les cannibales

$$\begin{aligned} \text{can}GD1 &\triangleq \\ &\wedge \text{can}G' = \text{can}G - 1 \\ &\wedge \text{can}D' = \text{can}D + 1 \end{aligned}$$

$$\begin{aligned} \text{can}GD2 &\triangleq \\ &\wedge \text{can}G' = \text{can}G - 2 \\ &\wedge \text{can}D' = \text{can}D + 2 \end{aligned}$$

$$\begin{aligned} \text{can}DG1 &\triangleq \\ &\wedge \text{can}D' = \text{can}D - 1 \\ &\wedge \text{can}G' = \text{can}G + 1 \end{aligned}$$

$$\begin{aligned} \text{can}DG2 &\triangleq \\ &\wedge \text{can}D' = \text{can}D - 2 \\ &\wedge \text{can}G' = \text{can}G + 2 \end{aligned}$$

$$\begin{aligned} \text{Init} &\triangleq \\ &\wedge \text{miss}G = \text{numMiss} \\ &\wedge \text{miss}D = 0 \\ &\wedge \text{can}G = \text{numCan} \\ &\wedge \text{can}D = 0 \\ &\wedge \text{barque} = \text{"G"} \end{aligned}$$

Bouger 1 missionnaire $G \rightarrow D$

$$\begin{aligned} \text{bougerM_GD} &\triangleq \\ &\wedge \text{barqueGD} \\ &\wedge \text{miss}G \geq 1 \\ &\wedge \text{missGD1} \\ &\wedge \text{UNCHANGED } \langle \text{can}G, \text{can}D \rangle \\ &\wedge \text{pasMiam'} \end{aligned}$$

Bouger 2 missionnaires $G \rightarrow D$

$$\begin{aligned} \text{bougerMM_GD} &\triangleq \\ &\wedge \text{barqueGD} \\ &\wedge \text{miss}G \geq 2 \end{aligned}$$

$\wedge missGD2$
 $\wedge \text{UNCHANGED } \langle canG, canD \rangle$
 $\wedge pasMiam'$

Bouger 1 cannibale $G \rightarrow D$
 $bougerC_GD \triangleq$
 $\wedge barqueGD$
 $\wedge canG \geq 1$
 $\wedge canGD1$
 $\wedge \text{UNCHANGED } \langle missG, missD \rangle$
 $\wedge pasMiam'$

Bouger 2 cannibales $G \rightarrow D$
 $bougerCC_GD \triangleq$
 $\wedge barqueGD$
 $\wedge canG \geq 2$
 $\wedge canGD2$
 $\wedge \text{UNCHANGED } \langle missG, missD \rangle$
 $\wedge pasMiam'$

Bouger 1 de chaque $G \rightarrow D$
 $bougerMC_GD \triangleq$
 $\wedge barqueGD$
 $\wedge missG \geq 1$
 $\wedge canG \geq 1$
 $\wedge missGD1$
 $\wedge canGD1$
 $\wedge pasMiam'$

Bouger 1 missionnaire $D \rightarrow G$
 $bougerM_DG \triangleq$
 $\wedge barqueDG$
 $\wedge missD \geq 1$
 $\wedge missDG1$
 $\wedge \text{UNCHANGED } \langle canG, canD \rangle$
 $\wedge pasMiam'$

Bouger 2 missionnaires $D \rightarrow G$
 $bougerMM_DG \triangleq$
 $\wedge barqueDG$
 $\wedge missD \geq 2$
 $\wedge missDG2$
 $\wedge \text{UNCHANGED } \langle canG, canD \rangle$
 $\wedge pasMiam'$

Bouger 1 cannibale $D \rightarrow G$

$$\begin{aligned}
bougerC_DG &\triangleq \\
&\wedge barqueDG \\
&\wedge canD \geq 1 \\
&\wedge canDG1 \\
&\wedge UNCHANGED \langle missG, missD \rangle \\
&\wedge pasMiam'
\end{aligned}$$

$$\begin{aligned}
&\text{Bouger 2 cannibales } D \rightarrow G \\
bougerCC_DG &\triangleq \\
&\wedge barqueDG \\
&\wedge canD \geq 2 \\
&\wedge canDG2 \\
&\wedge UNCHANGED \langle missG, missD \rangle \\
&\wedge pasMiam'
\end{aligned}$$

$$\begin{aligned}
&\text{Bouger 1 de chaque } D \rightarrow G \\
bougerMC_DG &\triangleq \\
&\wedge barqueDG \\
&\wedge missD \geq 1 \\
&\wedge canD \geq 1 \\
&\wedge missDG1 \\
&\wedge canDG1 \\
&\wedge pasMiam'
\end{aligned}$$

$$\begin{aligned}
Next &\triangleq \\
&\vee bougerM_GD \\
&\vee bougerMM_GD \\
&\vee bougerC_GD \\
&\vee bougerCC_GD \\
&\vee bougerMC_GD \\
&\vee bougerM_DG \\
&\vee bougerMM_DG \\
&\vee bougerC_DG \\
&\vee bougerCC_DG \\
&\vee bougerMC_DG
\end{aligned}$$

$$Spec \triangleq Init \wedge \Box[Next]_{\langle missG, missD, canG, canD, barque \rangle}$$